MERIT USER GUIDE

Importing NPI and CLIA records

How to load laboratories, ordering facilities and providers into MERIT's reporter registry from a spreadsheet, check the file before anything is saved, fix the rows MERIT cannot load, and clear the results that were waiting for them.

In one minute

1. Open **Reporters** → **Import NPI and CLIA records** and download the CSV template.
2. Fill in one row per laboratory, facility or provider: its `type`, name, and its CLIA number or NPI.
3. Click **Check the file**. Nothing is saved yet: MERIT lists the ready rows and explains every row it cannot load.
4. Click **Import**. Ready rows are added, or update the entry with the same CLIA number or NPI.
5. Fix the other rows and check again. Then release any results held as **unknown sender**.

Who can import: people with the *Edit the reporter registry* permission (intake clerks, supervisors and administrators). Everyone who can see people can search the registry. A practice file, `sourceCode/samples/registry_npi_clia_sample.csv`, comes with MERIT.

Companion guide: *MERIT User Guide - Daily Operations*. Interactive diagram: `docs/npi-clia-import_workflow.html`.

1. What the reporter registry is for

Every blood test result in MERIT names the **laboratory** that ran it and, usually, the **facility and provider** who ordered it. The reporter registry is MERIT's list of them, each identified by a national number:

Kind	Examples	Identified by	Why MERIT needs it
Laboratory	A reference laboratory, a hospital laboratory	CLIA number (required); NPI optional	Intake links every result to its laboratory by CLIA number. A result from a laboratory that is not registered is **held as unknown sender**.
Facility	A pediatric practice, a clinic, a hospital, a health center	NPI (organization) or CLIA number	Shows who ordered the test, for follow-up calls and the reporter timeliness report.
Provider	A physician, nurse practitioner or physician assistant	NPI (individual)	Links the ordering provider on each result, for confirmatory test and prenatal notifications.

2. NPI and CLIA numbers in brief

NPI: National Provider Identifier

`1245319599`

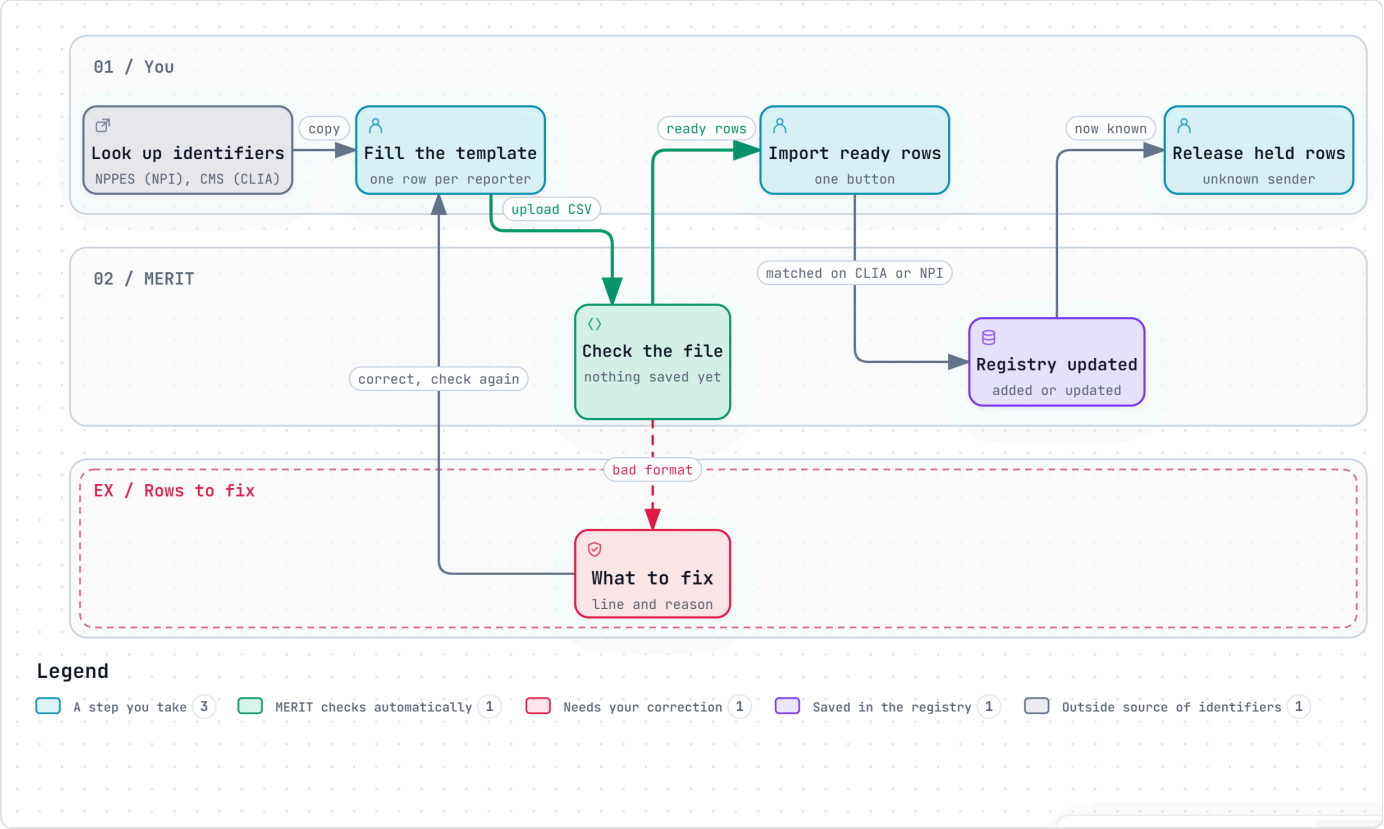
- **10 digits**, no letters.
- The **last digit is a check digit**. One wrong or swapped digit almost always fails it, so MERIT catches most typing mistakes.
- Individuals and organizations each have their own NPI.
- Look one up in the national NPPES NPI Registry (npiregistry.cms.hhs.gov).

CLIA number

`21D0123456`

- **Two digits, the letter D, then seven digits**: ten characters.
- Issued by CMS to every laboratory that tests human specimens.
- Laboratories print it on their reports; you can also look it up with CMS's CLIA laboratory search.
- Providers do not have CLIA numbers.

MERIT ignores spaces and hyphens in both (`21D-0123456` and `1245 319 599` are accepted) and accepts a lower-case `d`. It reads laboratory files the same way, so a result that carries a CLIA number or NPI written with spaces or hyphens still links to the registered entry.



The import workflow. Interactive version: `docs/npi-clia-import_workflow.html`.

3. Prepare the file

1. Open **Reporters** and click **Import NPI and CLIA records**.
2. Click **Download the CSV template**. It has the column names and one example of each kind.
3. Open it in Excel or another spreadsheet. Replace the examples with your rows: one row per laboratory, facility or provider.
4. Save as **CSV** (in Excel: *File → Save As → CSV UTF-8 (Comma delimited)*). Keep the first line, the column names, unchanged.

The columns

Column	What to put in it	Laboratory	Facility	Provider
`type`	`laboratory`, `facility` or `provider`	**Required**	**Required**	**Required**
`name`	Laboratory or facility name, as it should appear in MERIT (up to 120 characters)	**Required**	**Required**	Leave blank
`lastName`, `firstName`	The provider's name	Leave blank	Leave blank	**Last name required**
`credential`	MD, DO, NP, PA…			Optional
`npi`	10-digit NPI	Optional	**NPI or CLIA**	**Required**
`clia`	CLIA number, for example 21D0123456	**Required**	**NPI or CLIA**	**Must be blank**
`phone`	Main phone number, for follow-up calls	Optional	Optional	

An example

```
type,name,lastName,firstName,credential,npi,clia,phone
laboratory,Tidewater Clinical Laboratory,,,,,21D0456789,410-555-0170
facility,Patapsco Family Health Center,,,,1987654328,,410-555-0188
provider,,Okafor,Grace,MD,1543210987,,
```

> **Spreadsheet tips.** Keep NPIs as plain 10-digit numbers: if Excel shows `1.24532E+09`, widen the column or format it as Text before saving. Do not add total rows, notes or extra header lines. Up to 5,000 rows and 2 MB per file; split bigger lists.

4. Check the file

1. On the import page, under **Choose the file**, click **Choose File** and pick your CSV.
2. Click **Check the file**. **Nothing is saved yet.**
3. Read the summary: how many rows are **ready** (by kind) and how many **need attention**. Each row that needs attention is listed with its line number in the file and what to fix.

Check results for registry_npi_clia_sample.csv

4 row(s) ready (1 laboratories, 1 facilities, 2 providers) · **4** row(s) need attention.

Import 4 ready row(s)

Line	Name	What to fix
3	Allegany Regional Lab	CLIA number 21D04567 is not valid: it is two digits, the letter D, then seven digits (for example 21D0123456).
5	Severn Pediatric Associates	NPI 1765432100 is not valid: an NPI is 10 digits and its last digit is a check digit. Copy it again from NPPES.
8	Patel	A provider needs an NPI.
9	Okafor, Grace	Same NPI as line 6; only the first is loaded.

Checking the practice file: four rows ready, four to fix, each with its line number and what to fix.

What the messages mean

Message	What to do
NPI … is not valid: an NPI is 10 digits and its last digit is a check digit	A digit is wrong or missing. Copy the NPI again from the NPPES NPI Registry.
CLIA number … is not valid: it is two digits, the letter D, then seven digits	Usually a digit is missing or the letter is not D. Copy it from the laboratory report or the CMS search.
A laboratory needs its CLIA number	Fill in `clia`. If you only have an NPI, the row is probably a facility: change `type`.
A facility needs an NPI or a CLIA number	Fill in one of them.
A provider needs an NPI / a last name	Fill in the missing column. Providers without an NPI cannot be registered.
Providers have no CLIA number; leave clia blank	Clear the `clia` cell, or change `type` to laboratory.
Same NPI (or CLIA number) as line …; only the first is loaded	The file lists the same reporter twice. A laboratory and a facility with the same CLIA number or NPI count as the same organization, whatever the `type` column says. Delete the duplicate row, or keep it if the first is correct.
Type must be laboratory, facility or provider	Check the spelling in the `type` column.
… is longer than … characters	Shorten the name, credential or phone.

5. Import

1. When the ready rows look right, click **Import *n* ready row(s)**.
2. MERIT shows how many entries were **added** and how many **updated**, and lists the rows it did not load.
3. Fix those rows in your spreadsheet. Save, then check and import the file again. Rows already loaded are simply updated again, so re-importing the whole file is safe.

How MERIT decides "add" or "update"

- **Laboratories and facilities** are matched on their CLIA number first, then their NPI. A match is updated with the new name and phone; otherwise a new entry is added.
- **Providers** are matched on their NPI.
- **Blank cells never erase** what is stored. To remove a phone number or end a reporter, ask your administrator.

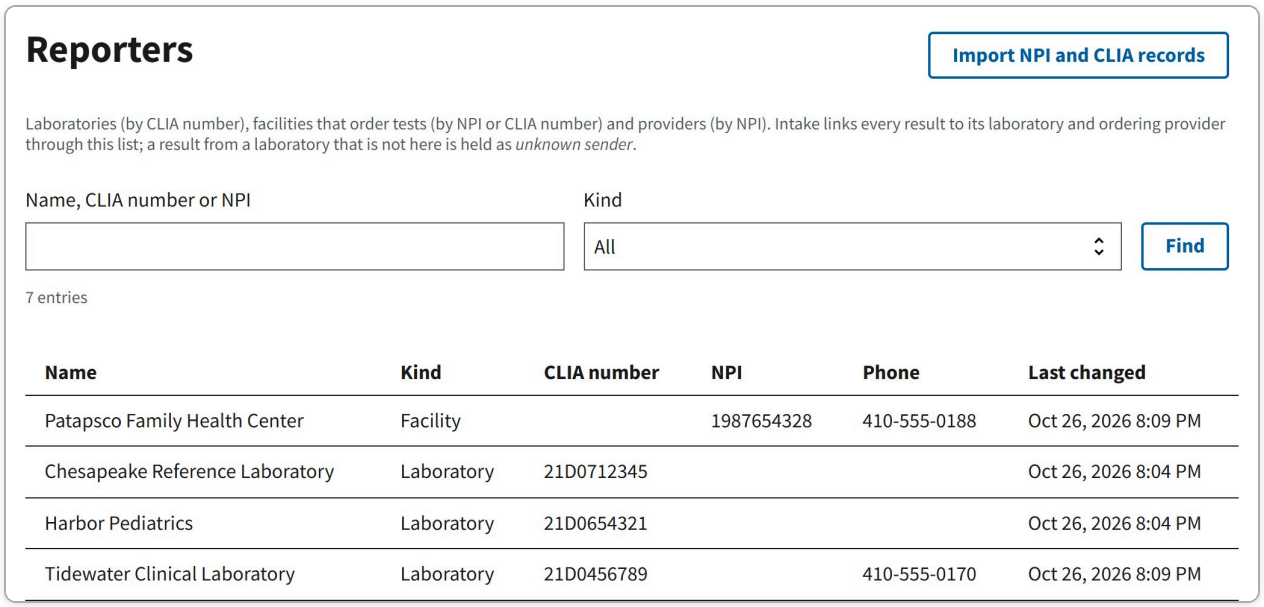
> **The check is valid for 30 minutes.** If you wait longer, MERIT asks you to check the file again before importing. That way the rows imported are always the rows you just reviewed.

6. After the import: release the held results

Results that arrived before their laboratory was registered are waiting on **Held records** with the cause **unknown sender**.

1. Open **Held records** and click the **Unknown sender** group for that laboratory.
2. Tick the rows (or the box at the top of the table for all of them) and click **Release selected**.
3. MERIT checks them again. Now that the laboratory is known they load, or wait for another cause such as match review.

To confirm an entry, search for it on **Reporters** by name, CLIA number or NPI.

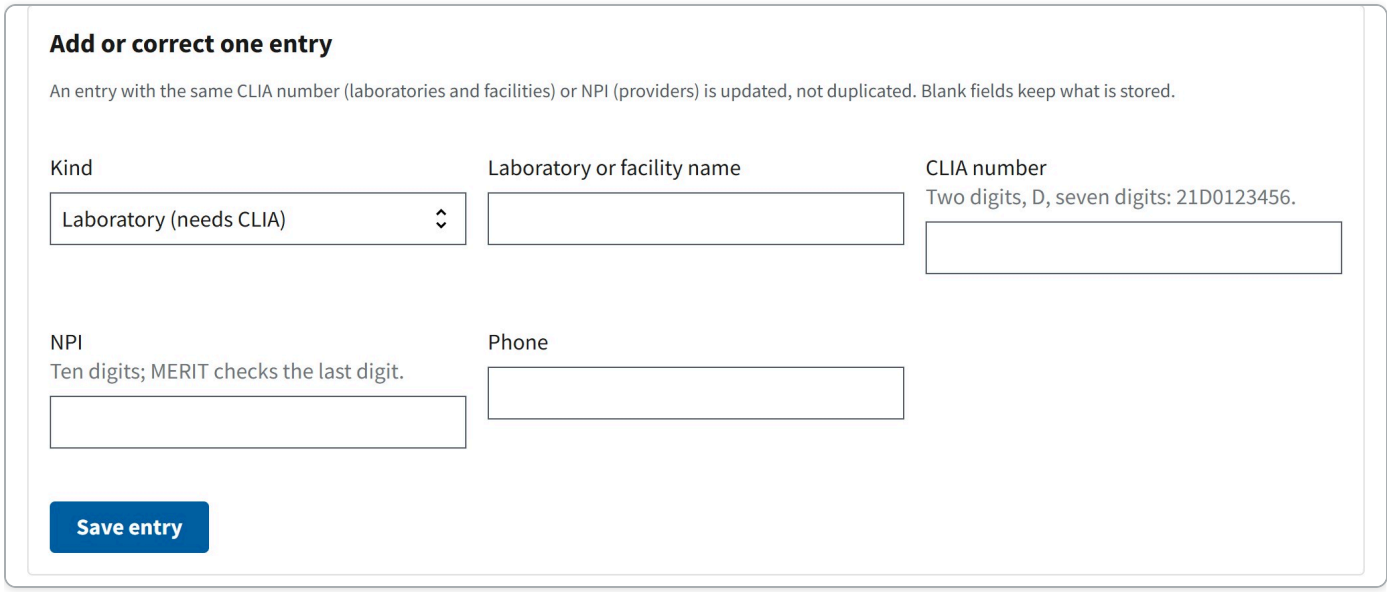


Reporters: search by name, CLIA number or NPI, and filter by kind. Entries added by the import appear here at once.

7. Adding or correcting one entry

For a single new laboratory or a corrected name you do not need a file.

1. Open **Reporters** and search for the reporter first.
2. Under **Add or correct one entry** (below the list), choose the **Kind**. The form shows only that kind's fields; fill in the name and identifiers.
3. Click **Save entry**. The same rules apply as for a file: identifiers are checked, and an entry with the same CLIA number or NPI is updated.



Add or correct one entry. A laboratory needs its name and CLIA number; a facility its name and an NPI or CLIA number; a provider a last name and NPI. The form shows the fields for the kind you choose.

8. Good practice

- **Register a laboratory before its first file arrives** when a new laboratory starts reporting. Ask your administrator to set up its intake profile at the same time.
- **Once a month**, open **Reports** → **Reporter timeliness and completeness** to find reporters that send late or leave fields blank.
- **Keep one master spreadsheet** of your reporters and re-import it after big changes, such as a hospital system merger. Re-importing updates; it never duplicates.
- **Use the names people recognize**, for example "Harbor Pediatrics - Dundalk" for one branch of a practice.

9. Common questions

I imported the same file twice. Do I now have duplicates?
No. The second import updated the entries the first one added.

A laboratory has two CLIA numbers (two sites).
Register each site as its own laboratory row, with its own CLIA number and a name that tells them apart.

The NPI is right but MERIT says the check digit is wrong.
Check for a dropped or extra digit, or a cell that the spreadsheet changed (scientific notation, removed leading characters). Copy it again from the NPPES NPI Registry.

Can I import people's test results this way?
No. This import is only for reporters. Results come in on **Intake** (see the Daily Operations guide).

The Import button is missing.
Your role or profile does not include *Edit the reporter registry*. Ask your supervisor or administrator.

MERIT user guide: importing NPI and CLIA records. The examples use made-up reporters with identifiers that pass the format checks.